

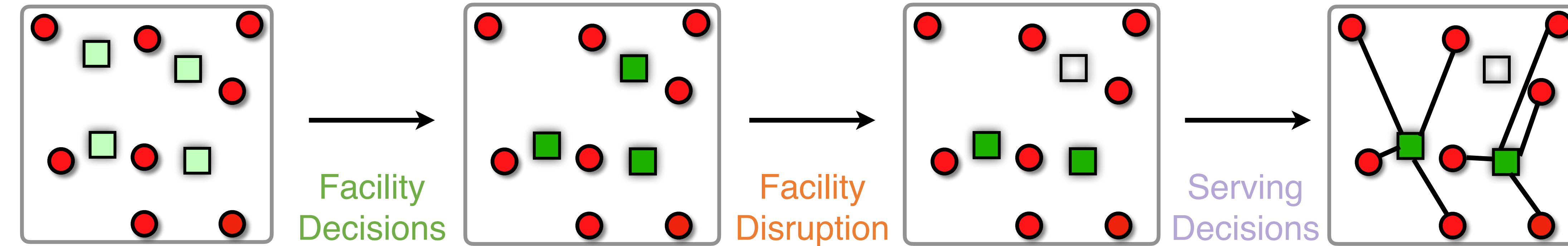
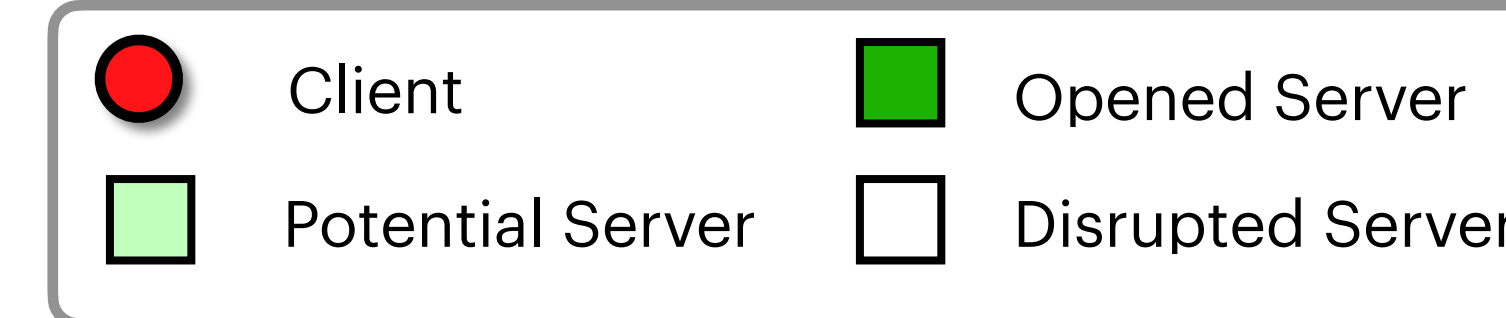
Neur2RO: Neural Two-Stage Robust Optimization

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Two-Stage Robust Facility Location Problem

Objective: Determine the optimal set of facilities to open under the worst-case facility disruption.



Optimization Formulation

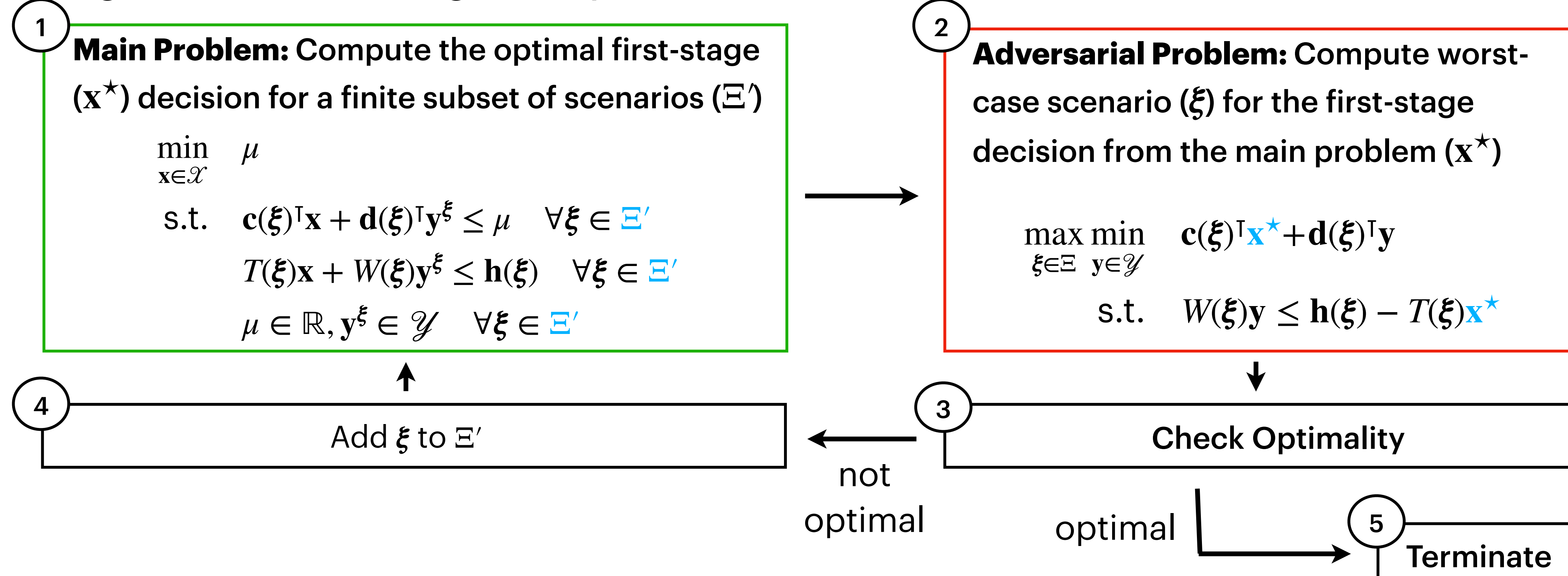
Goal: Determine optimal **first-stage decisions** that minimize the sum of the **first-stage cost** and **second-stage cost** for the **worst-case scenario**.

Challenge: Solving the nested optimization problem is intractable and specialized algorithms only exist for limited classes of problems.

$$\min_{\mathbf{x} \in \mathcal{X}} \underbrace{\max_{\xi \in \Xi}}_{\text{Worst-Case Uncertainty}} \underbrace{\min_{\mathbf{y} \in \mathcal{Y}}}_{\text{Second-Stage Decisions}} \underbrace{\mathbf{c}(\xi)^T \mathbf{x} + \mathbf{d}(\xi)^T \mathbf{y}}_{\text{First-Stage Cost}} \quad \text{s.t.} \quad T(\xi)\mathbf{x} + W(\xi)\mathbf{y} \leq \mathbf{h}(\xi)$$

Solving 2RO Problems

• Column-and-Constraint Generation is a general purpose, but inefficient algorithm for solving 2RO problems.



Machine Learning Methodology

ML Solution: Replace the challenging exact optimization problems with a neural network approximation by learning the value function of the innermost optimization problem.

Prediction Task

$$NN_{\Theta}(\mathbf{x}, \xi) \approx \min_{\mathbf{y} \in \mathcal{Y}} \{ \mathbf{c}(\xi)^T \mathbf{x} + \mathbf{d}(\xi)^T \mathbf{y} : W(\xi)\mathbf{y} \leq \mathbf{h}(\xi) - T(\xi)\mathbf{x} \}$$

Main Problem Approximation

$$\min_{\mathbf{x} \in \mathcal{X}, \mathbf{y} \in \mathcal{Y}, \xi_a \in \Xi} \mathbf{c}(\xi_a)^T \mathbf{x} + \mathbf{d}(\xi_a)^T \mathbf{y} \quad \text{s.t.} \quad \begin{cases} W(\xi_a)\mathbf{y} + T(\xi_a)\mathbf{x} \leq \mathbf{h}(\xi_a) \\ \xi_a \in \arg \max_{\xi \in \Xi'} \{ NN_{\Theta}(\mathbf{x}, \xi) \} \end{cases}$$

Use network as a worst-case scenario classifier in optimization

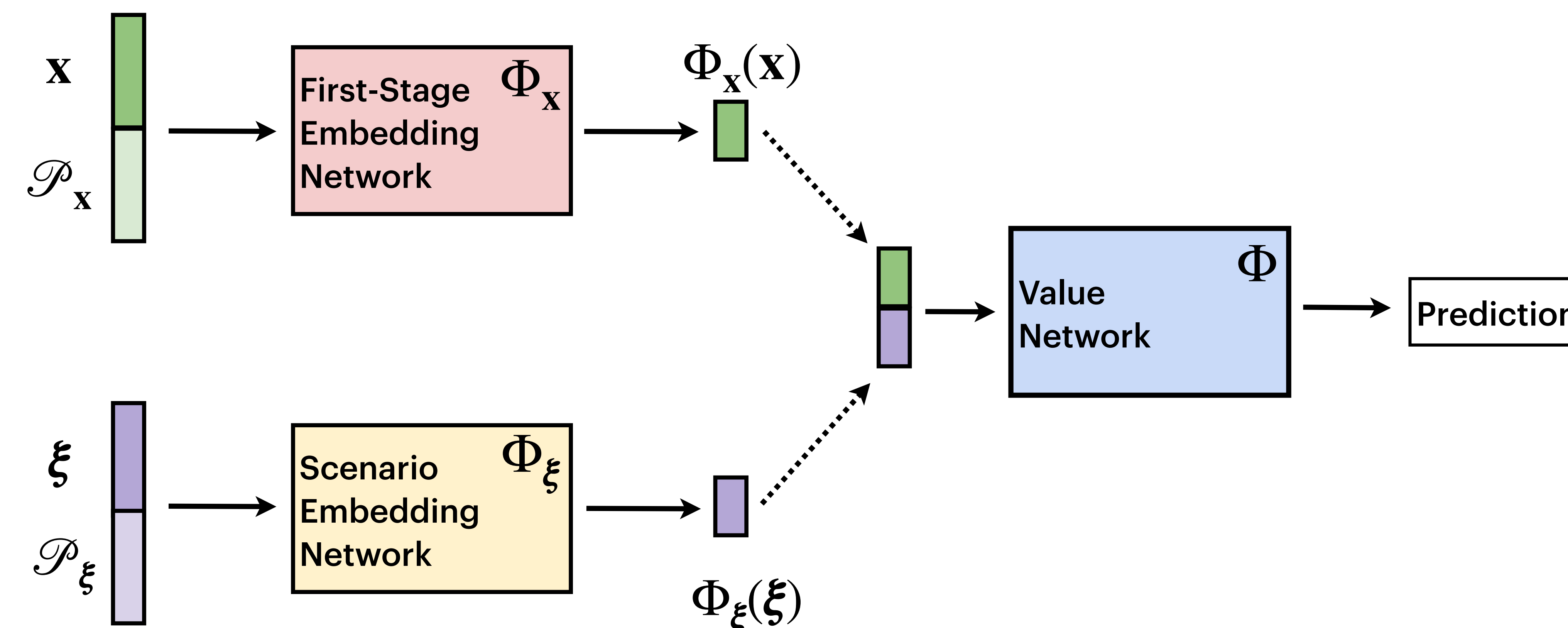
Use coefficients of problem for identified worst-case scenario

Adversarial Problem Approximation

$$\max_{\xi \in \Xi} NN_{\Theta}(\mathbf{x}^*, \xi)$$

Requires finding the input (scenario) that leads to the maximum output for a given first-stage decision

Architecture

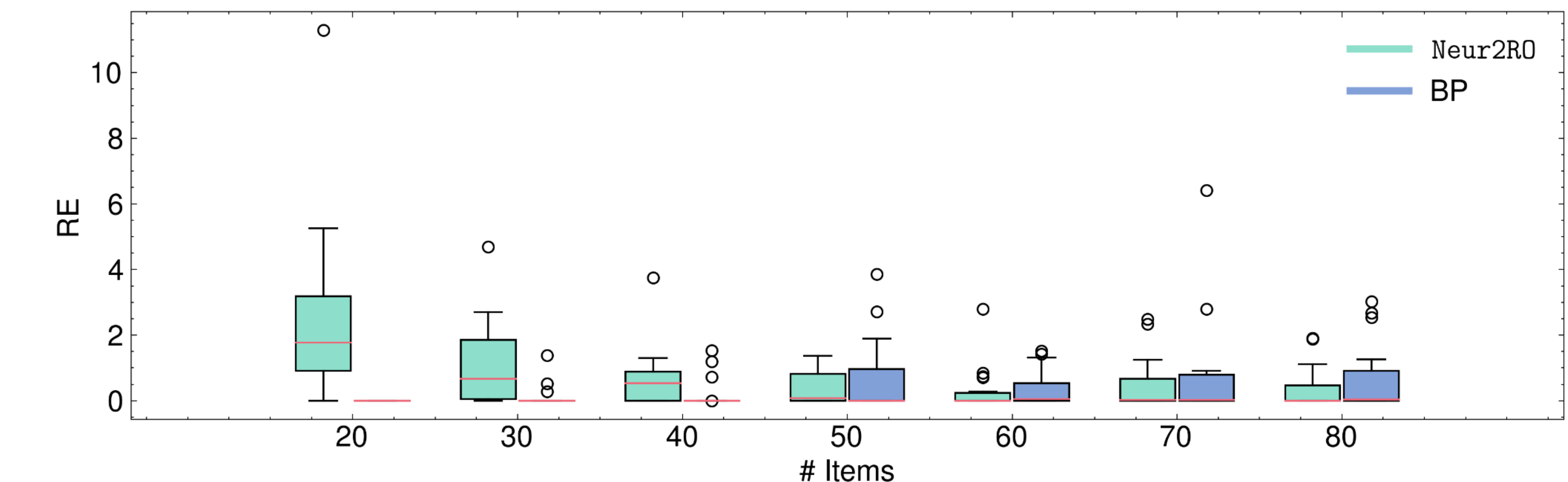


- Architecture enables efficient optimization as only the **Value Network** is added for each new scenario in the main problem.
- Only one of the **Scenario/Decision Embedding Networks** need to be added to optimization model for main/adversarial problem.

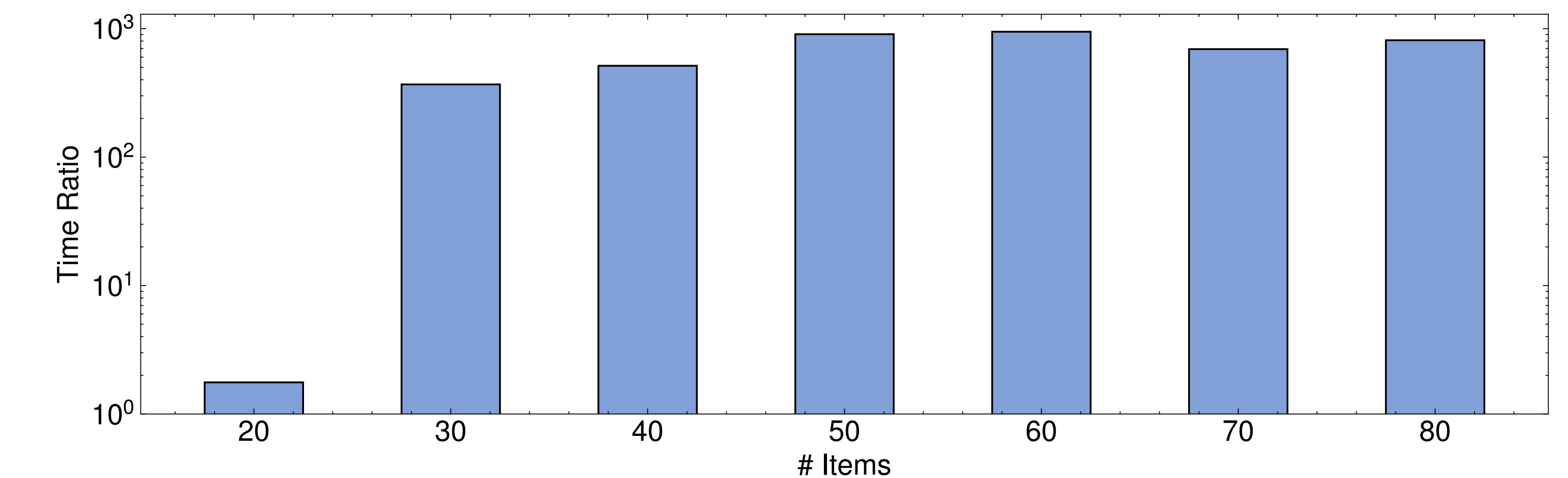
Experimental Results

Knapsack Problem

Relative error to best known solution branch-and-price (lower is better)

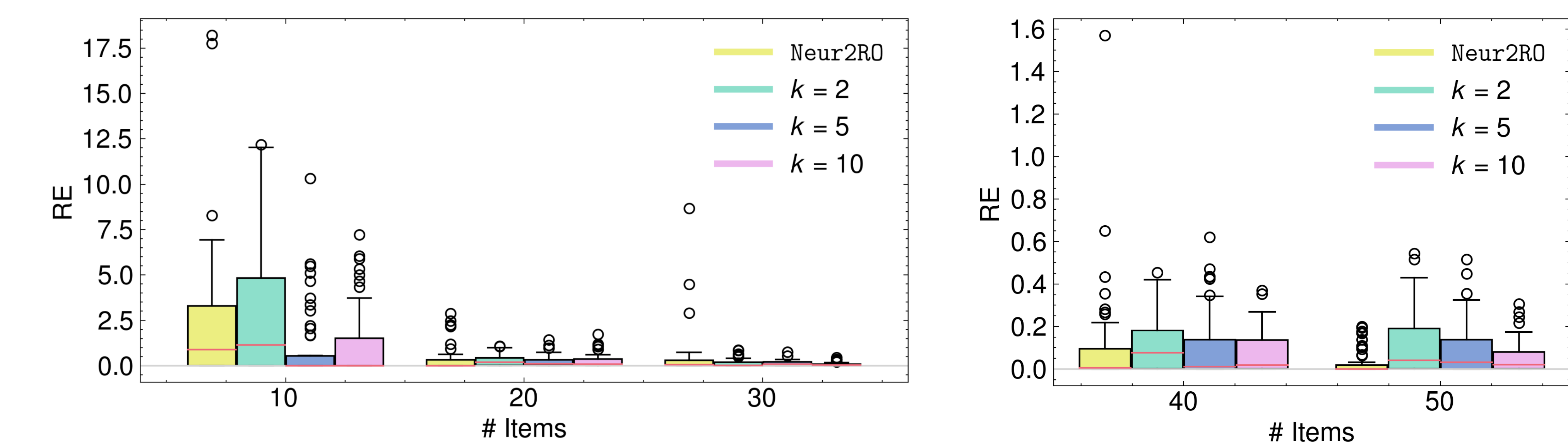


Reduction factor in solving time over branch-and-price (higher is better)

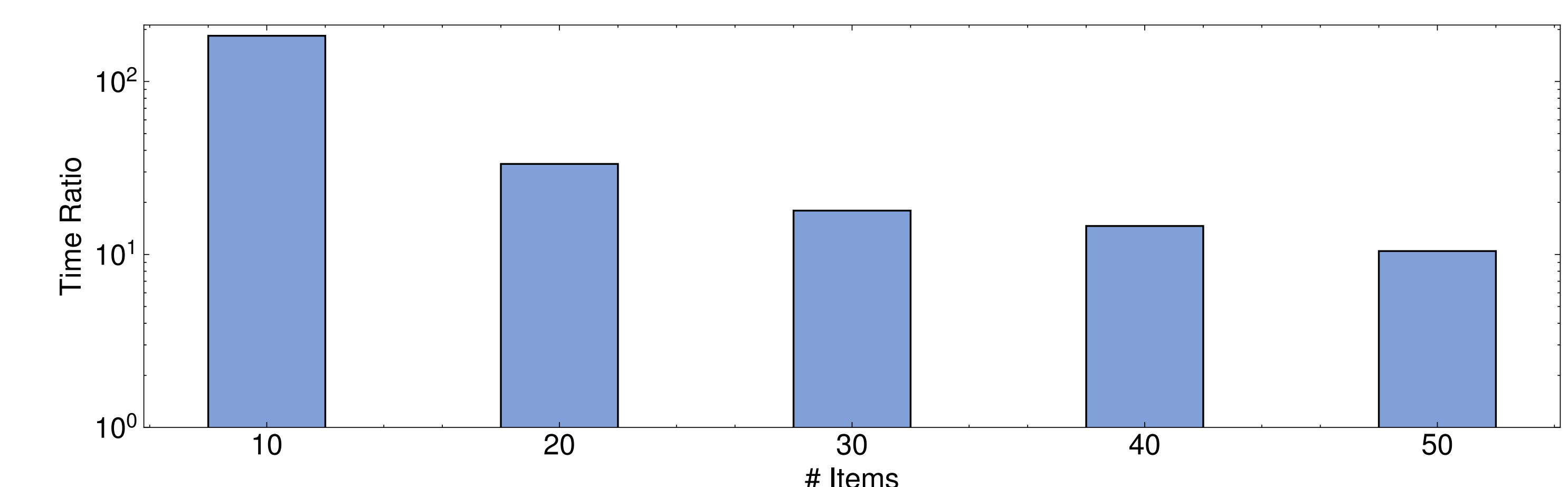


Capital Budgeting Problem

Relative error to best known solution over k-adaptability (lower is better)



Reduction factor in solving time over k-adaptability (higher is better)



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